

Gemini Near Infra-Red Imager and Wavefront Sensor



Near IR Imager (NIRI) and On-Instrument Wavefront Sensor (OIWFS)—Software Installation and Release Notes

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This document describes the procedure for building and installing the NIRI/OIWFS software, the NIRI engineering software, and the Near Infra-Red OIWFS software, release 1.00.

1 Introduction

1.1 Purpose

This document describes the procedure for building and installing the NIRI CC and OIWFS software. It also describes version-specific features of this release.

This package provides all of the software that is necessary to control NIRI or the OIWFS according to gemini specifications. The higher level CAD/CAR/APPLY interface is provided by niri_1.00 which is based on the CICS. The CICS and the higher-level interface software was written by Steven Beard.

The low-level engineering package is part of the software distribution. It provides the functions that are necessary to operate the NIRI and OIWFS mechanisms. It may also be used as a stand-alone package which provides all the tools that are necessary to interface to the NIRI CC and OIWFS hardware, to verify that the hardware is working correctly, and to calibrate the hardware. When used as a standalone package, it does not satisfy all of the gemini interface requirements.

The intended audience of this document is:

- the NIRI reviewers;
- the developers of the instruments which will be using the NIRI OIWFS.

1.2 Definitions, Acronyms and Abbreviations

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

1.3 Stylistic Conventions

References to documents in Section 1.5, “References,” on page 2 of “Near IR Imager (NIRI)—Definitions, Abbreviations, and References” are given like this; [1].

1.4 References

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

2 Release Notes

This is NIRI software 1.00, which is intended to be the final software release of the University of Hawaii software.

The following features are missing or broken and are expected to be fixed by the Gemini telescope developers.

- Some hardware subsystems (window cover, air ionizer) are not supported;
- The World Coordinate System is not supported;
- Communications with the detector controller need further work;
- Health problems cannot all be traced to alarms;

2.1 The 1.00 release (2000-06-27)

Currently under development.

- Feature Freeze;
- WCS code now added (though WCS support still not present);
- A&G Interface now conforms fully to the ICD’s;
- Some of the A&G interface and time-handling software was changed, to conform to the A&G system, as implemented;
- The documentation has been finalized.

2.2 The 1.00 alpha 6 release (1999-11-11)

The alpha 6 release includes the following changes:

- Fixed many small problems with the Engineering User interface;
- Fixed some problems with the OMS driver which occurred when the requested step count was too large;
- Added start-up scripts for OIWFS only mode;
- The “INVALID” position warning and the move-verification errors are now triggered by the same tolerances;
- The A&G interface now conforms better to the ICD’s;

- The directory structure has been significantly rearranged, to be more consistent;
 - The CICSCC directory is renamed to CC;
 - The CICSWFS directory has been renamed to WFS;
 - The gnaac-interface directory is now a complete EPICS-application tree (the DC directory);
 - The sequencer functions have been moved into an IS directory;
 - The GNAAC simulator directory has been removed;
- The startup scripts have been rewritten to be easier to modify;
- WfsBeam is now implemented;
- The directory structure and configuration files are now finalized.

2.3 The 1.00 alpha 5 release (1999-10-23)

The alpha 5 release includes the following changes:

- Restructured CAD files to make the logic clearer;
- The A&G interface is now functional.

2.4 Release V003, Releases 1.00 alpha 1 - 1.00 alpha 4

These were not publicly released. They involved extensive restructuring of the CICS-derived code to make it easier to modify. They also included changes to support communications with the detector controller; better support for the Hall effect sensors; temperature control; interlocks; temperature-dependent records; support for the full range of NIRI mechanisms; and many other smaller changes.

2.5 Release V002

Mostly bug fixes and changes in the way that configuration is handled. There is now a sequencer control screen.

2.6 Release V001

This is the initial release and was not widely distributed.

3 Hardware configuration

3.1 VME Boards

All boards should be set as specified in [30], except as described below.

The address of the Oregon Microsystems boards must be set to 0xFC80 (first board) and 0xFC90 (second board). The address of the XYCOM 490 board must be set to 0xE000. (See your Oregon Microsystems and Xycom documentation). Interrupt vector 0xE0 must be available for use by the Xycom board.

3.2 Omega Temperature controllers

The Omega temperature controllers should be set to 1200 baud, using the front panel commands.

3.3 MV167 Real time clock

Unfortunately, some of the EPICS device support (e.g., the steppermotor record) assumes a 60 Hz clock, which can cause occasional, intermittent problems; since the NIRI/OIWFS software no longer requires the steppermotor record, it may no longer be necessary to set the real time clock, but it is recommended as a precaution

The following instructions will fix the problem. Warning: These instructions are only used for the MV167, which is the gemini standard CPU for instrument systems. *Do not follow these instructions on any other system!* At the VxWorks prompt:

```
m 0xffffc00fe
```

You will see 0xffffc00fe:, followed by a 4 digit hexadecimal number, followed by a dash. Record it for later reference. Enter:

```
21ee
```

You will see 0xffffc0100, followed by a four digit hexadecimal number, followed by a dash. Enter a period:

```
.
```

To undo these changes, repeat the procedure, except that you must enter the number that you recorded earlier, instead of 21ee.

4 Prerequisites

This software was developed under Solaris 2.5.1 and epics3.12.2GEM5 / VxWorks 5.2. I do not guarantee that it will work correctly with any other operating system release or EPICS release. At this time no customization to the operating system is necessary. Minor changes to EPICS may be necessary, as described below.

In order to run the CICS/NIRI interface to the GNAAC detector controller, a current version of the GNAAC detector controller software must have already been installed and running on a separate IOC. (The GNAAC simulator, is no longer included with the software, because it is very out of date.)

In order to build and to run the CICS/NIRI software, the Gemini slalib, Gemini astlib, and Gemini timelib are all required.¹

Also required:

- gunzip.

1. These packages are available via ftp from <ftp://ftp.gemini.edu/pub/epics/gemLibs.tar.gz>

4.1 VxWorks routing bug

There is a bug in VxWorks that sometimes causes communications problems between the instrument sequencer and the detector controller software (running on a separate CPU).

Execute a `routeShow` command, from the VxWorks prompt. This will show your routing table. Typically, you will see two entries

destination	gateway
0.0.0.0	xxx.xxx.xxx.rrr ^a
xxx.xxx.xxx.0 ^b	xxx.xxx.xxx.yyy ^c

- a. The address of your router.
- b. Your local subnetwork address.
- c. Your host address.

On some systems, you may see the following:

destination	gateway
0.0.0.0	xxx.xxx.xxx.rrr ^a
xxx.xxx.xxx.0 ^b	xxx.xxx.xxx.yyy ^c
xxx.xxx.xxx.0	xxx.xxx.xxx.rrr

- a. The address of your router.
- b. Your local subnetwork address.
- c. Your host address.

If you see this, you should edit your `$local.vws` file, located in `$EPICS/extensions/src/uae/templates/startup`. There will normally be a line in this file of the form:

```
routeAdd "0", "xxx.xxx.xxx.rrr"
```

where `xxx.xxx.xxx.rrr` is the address of your router.

After this line, add the command:

```
routeDelete "xxx.xxx.xxx.0", "xxx.xxx.xxx.rrr"
```

where `xxx.xxx.xxx.0` is the address of your local subnetwork and `xxx.xxx.xxx.rrr` is the address of your router.

If you are unable to modify your epics distribution, you may add this command to your VxWorks startup script instead.

4.2 Bug fix for e2sr (pre-GEM5)

Important: There is a bug in the `e2sr` command, which is part of the EPICS distribution (fixed in GEM5). This bug will cause `e2sr` to crash, if a variable is set to an empty value. It is not possible to compile the software, unless the bug is patched.

If an version of EPICS preceding GEM5 is used², the following patch should be applied:

Edit `edifParser()`, in `$EPICS/extensions/src/edif/edifLib.c`, by replacing

```
while (*in != EOS) {  
  
    ...  
  
}
```

with

```
if (in != NULL) {  
    while (*in != EOS) {  
  
        ...  
  
    }  
  
}
```

5 Building, Loading, and Using the NIRI Low-Level Engineering Software

The `niriEng` package is a minimal package which is designed to run the NIRI and OIWFS hardware, and to simulate the hardware. It does not satisfy the Gemini interface requirements. It does provide all functions which are necessary to calibrate and debug the NIRI hardware.

Note: the full NIRI/CICS package, described below in “Building, Loading, and Using the CICS/NIRI/OIWFS software” on page 10, uses the `niriEng` package³ for its low-level functions. All functions which are provided by the `niriEng` package are available in the CICS/NIRI package; there are three reasons to use the `niriEng` package directly:

- It loads a *lot* faster
- It compiles a *lot* faster
- It uses less memory (probably not a real problem)

5.1 Obtaining the Software

Ftp to `gemftp.ifa.hawaii.edu`, and download

```
/pub/gemini_IFA/niri/source/niri_1.00.tar.gz
```

2. This is not to imply that releases of EPICS other than GEM5 are supported!

3. The previous release of this package was called `niriCC`

For instruments other than NIRI, there will also be a file:

```
/pub/gemini_IFA/niri/source/XXXX_1.00.gz
```

Where XXXX will be the name of the instrument, e.g., “gnirs.” Get this file in addition to the standard niri file. This file contains hardware parameters which are unique for each mechanism that is built.

5.2 Unpacking the Software

Execute the following commands:

```
mkdir niri_1.00
cd niri_1.00
cp (...) /niri_1.00.tar.gz .
gunzip < niri_1.00.tar.gz | tar -xvf -
```

For instrument XXXX (other than niri), after executing the previous commands, execute:

```
cp (...) /niri_1.00_XXXX_00.tar.gz .
gunzip < niri_1.00_XXXX_00.tar.gz | tar -xvf -
```

You only need the ENG subdirectory. The use of the remaining contents of this directory is described below, in “Building, Loading, and Using the CICS/NIRI/OIWFS software” on page 10.

5.3 Configuring a standard EPICS system

Previous releases of this software required several changes to the EPICS distribution. This package now uses a modified version of the motor record, and does not need to replace the stepper motor record.

5.4 Configuration

Create a `niriLogin` file; the file `niriLogin_SAMPLE` is provided as an example. You will need to edit this file to set the path to your installed NIRI software, and to set your prefix (or prefixes).⁴

Warning: The instrument prefix, e.g., `niri:`, must contain at most five characters, including the final colon. The status-and-alarm-database prefix, e.g., `niri:` or `niri::`, must contain at most six characters, including the final colon. The instrument name, e.g., `NIRI`, must contain at most four characters.

You should not need to change any other files. If you do need to do so, it should be reported as a bug.

The `csh/tcsh` configuration file “`niriLogin`” sets up various environment variables and your path. You must source `niriLogin` before building or running any of the niri software. It might be wise to source “`niriLogin`” in your `.login` or `.cshrc` file:

```
source niriLogin
```

4. `NiriLogin` assumes that you are using `csh`, `tcsh`, or some other shell that is compatible with the `csh/tcsh` syntax. If not, you will need to make an equivalent file for your shell, using `niriLogin` as a starting point.

The variables which must be defined in niriLogin are:

TABLE 1.

Variable	Default value	Purpose	Change? ^a
NIRI_ENGDIR	/gemini/niri/ENG	The path to the NIRI engineering source directory.	Always
NIRI_PREFIX	niri:	This prefix is prepended to all records. It should be four or fewer alphanumeric characters, followed by a colon.	If needed
NIRI_NAME	NIRI	This is used to generate a description of various records. It should be four or fewer alphanumeric characters.	If needed
NIRI_TMPSCAN	“Passive”	This is used by the temperature interlock, which is described in [6], Section 2–4.2.1, “Monitoring an external record.”, on page 10. Set this to a scan rate, e.g., “1 second”.	If needed
NIRI_TMPVAR	0	This is used by the temperature interlock, which is described in [6], Section 2–4.2.1, “Monitoring an external record.”, on page 10. Set this to a record field, e.g., niri:cc:tmpvar.	If needed
NIRI_TMPHB	0	This is used by the temperature interlock, which is described in [6], Section 2–4.2.1, “Monitoring an external record.”, on page 10. Set this to a record field, e.g., niri:cc:heartBeat.	If needed
NIRI_TARGETS	mv167	Which EPICS-supported hardware platform ^b .	Not recommended

- a. Items which are marked “Not recommended” will have unexpected results, unless the niri software is modified accordingly. Details are outside the scope of this document.
- b. The mv167 is the only supported architecture at this time.

5.4.1 Initial Setup

In the \$NIRI_ENGDIR directory, type:

```
./niriSetup
```

You will get a few warning messages, e.g., that “.cvsignore” does not exist. These are harmless and can be ignored.

5.4.2 Environment subsystem

The environment simulator is now working. *It is no longer necessary to disable it if the appropriate supporting hardware is not available.*

5.5 Building the software

In the \$NIRI_ENGDIR directory, type

```
gmake
```

This should build the system. The software should compile without warnings, *except*

- All C modules will warn “‘rcsid’ defined but not used.”
- The SNL code compiler (*.stpp files) generates a lot of warnings about undefined variables.

5.6 Making dependency list

Switch to the source directory

```
cd $NIRI_ENGDIR/src
```

Make the dependency list

```
gmake depends
```

This will insure that all files will be remade if the record configuration files or C header files are changed.

5.7 Loading the NIRI CC and OIWFS control software or simulator

Connect to your IOC and set the start-up script to:

- (...) /bin/mv167/startupSim (simulator),
- (...) /bin/mv167/startup (control software),
- (...) /bin/mv167/startupWfsSim (OIWFS simulator), or
- (...) /bin/mv167/startupWfs (OIWFS control software).

where (...) represents the full, absolute path of the niri engineering directory. Then reboot the IOC. The IOC should reboot without any errors. At this time, you should be running the software that you selected.

5.8 Running the diagnostic software

- Make sure that NIRI_ENGDIR is defined.
- Make sure that \$NIRI_ENGDIR/bin/solaris is in your path

Type:

```
niriEng
```

This should start up the niri low-level engineering screen. Note: This screen is designed to control the whole NIRI system. If it is used to control the OIWFS system, non-OIWFS control widgets will be white. Separate screens to control the OIWFS only are not provided.

The low level engineering screens are documented in [5].

5.9 Cleaning and rebuilding

The set up and installation process generates a number of files automatically. To insure that the old files are removed before building the system with a new configuration, type:

```
gmake reset
```

Do not use

```
gmake clean
```

because this fails to remove some files which should be removed, and remove some files which should not be removed. To restore the system to its original state, type:

```
gmake resetall.
```

6 Building, Loading, and Using the CICS/NIRI/OIWFS software

The niri package is designed to run the NIRI and OIWFS hardware, and to simulate the hardware. This package is intended to satisfy all Gemini interface requirements for a science instrument. It provides all functions which are necessary to calibrate and debug the NIRI hardware.

Note: the full NIRI/CICS package uses the niriEng package (described above in “Building, Loading, and Using the NIRI Low-Level Engineering Software” on page 6) for its low-level functions.

6.1 Obtaining the Software

Ftp to gemftp.ifa.hawaii.edu, and download

```
/pub/gemini_IFA/niri/source/niri_1.00.tar.gz
```

6.2 Unpacking the Software

Execute the following commands:

```
mkdir niri_1.00
cd niri_1.00
cp (...)niri_1.00.tar.gz .
gunzip < niri_1.00.tar.gz | tar -xvf -
```

For instrument XXXX (other than niri), after executing the previous commands, execute:

```
cp (...)niri_1.00_XXXX_00.tar.gz .
gunzip < niri_1.00_XXXX.tar.gz | tar -xvf -
```

6.3 Configuring a standard EPICS system

It is no longer necessary to modify your EPICS distribution. This package now uses a modified version of the motor record, and does not need to replace the stepper motor record.

However, the address of the Oregon Microsystems board must be set to 0xFC80 and the address of the XYCOM 490 board must be set to 0xE000. (See your Oregon Microsystems and Xycom documentation). Note also that interrupt vector 0xE0 must be available for use by the Xycom board.

6.4 Configuration

Create a niriLogin file; the file niriLogin_SAMPLE is provided as an example. You will need to edit this file to set the path to your installed NIRI software, and to set your prefix (or prefixes).⁵

Warning: The instrument prefix and detector controller prefix, e.g., niri:, must contain at most five characters, including the final colon. The status-and-alarm-database prefix and detector-controller status-and-alarm-database prefix, e.g., niri: or niri::, must contain

5. NiriLogin assumes that you are using csh, tcsh, or some other shell that is compatible with the csh/tcsh syntax. If not, you will need to make an equivalent file for your shell, using niriLogin as a starting point.

at most five characters, including the final colon. The instrument name, e.g., **NIRI**, should contain at most four characters.

The variables which must be defined in niriLogin are:

TABLE 2.

Variables

Variable	Default value	Purpose	Change ^a
NIRI_DIR	/gemini/niri	The path to the NIRI software source directory.	Always
NIRI_DCPKGDIR	/gemini/gnaac/gnaacSrc	The path to the NIRI detector-controller (GNAAC) source directory.	Always
NIRI_SLALIBDIR	/gemini/slalib	This is the path to the installed Gemini slalib package.	Always
NIRI_TIMELIBDIR	/gemini/timelib	This is the path to the installed Gemini timelib package.	Always
NIRI_ASTLIBDIR	/gemini/astlib	This is the path to the installed Gemini astlib package.	Always
NIRI_PREFIX	niri:	This prefix is prepended to all records. It should be four or fewer alphanumeric characters, followed by a colon.	If needed
NIRI_DCPREFIX	niri:	When accessing records in the detector controller, this prefix will be prepended to all records. It should be four or fewer alphanumeric characters, followed by a colon. It should not contain the dc: prefix, which is normally added to detector-controller records.	If needed
NIRI_SADPREFIX	\${NIRI_PREFIX}	This prefix is prepended to all records in the status and alarm database. It should be four or fewer alphanumeric characters, followed by one or two colons.	If needed
NIRI_DCSADPREFIX	\${NIRI_DCPREFIX}	When accessing records in the detector controller status and alarm database, this prefix will be prepended to all records. It should be four or fewer alphanumeric characters, followed by a colon. It should not contain the dc: prefix, which is normally added to detector-controller records.	If needed
NIRI_NAME	NIRI	This is used to generate a description of various records. It should be four or fewer alphanumeric characters.	If needed
NIRI_TMPSCAN	“Passive”	This is used by the temperature interlock, which is described in [6], Section 2–4.2.1, “Monitoring an external record.”, on page 10. Set this to a scan rate, e.g., “1 second”.	If needed

TABLE 2. Variables

Variable	Default value	Purpose	Change ^a
NIRI_TMPVAR	0	This is used by the temperature interlock, which is described in [6], Section 2–4.2.1, "Monitoring an external record.", on page 10. Set this to a record field, e.g., niri:cc:tmpvar.	If needed
NIRI_TMPHB	0	This is used by the temperature interlock, which is described in [6], Section 2–4.2.1, "Monitoring an external record.", on page 10. Set this to a record field, e.g., niri:cc:heartBeat.	If needed
NIRI_USEBANCOMM	0	If this is set to 1, then the Bancomm clock board is not initialized. This is useful for debugging the system when the timebus is not available ^b .	For debugging only
NIRI_USEVXTIME	1	If this is set to 0, then the internal system clock is used, instead of the timebus. This is useful for debugging the system when the timebus is not available ^c .	For debugging only
NIRI_AGPREFIX	ag:	This is the top-level prefix of the A&G system. The default value points to a simulation record, to avoid writing to the real A&G inadvertently. This may be set to aid in \${NIRI_PREFIX} debugging the system ^d .	For debugging only
NIRI_LIBDIR	\${NIRI_DIR}/LIB	The path to the NIRI shared source directory.	Not recommended
NIRI_ENGDIR	\${NIRI_DIR}/ENG	The path to the NIRI engineering source directory.	Not recommended
NIRI_CCDIR	\${NIRI_DIR}/CC	The path to the NIRI public components controller interface directory	Not recommended
NIRI_WFSDIR	\${NIRI_DIR}/WFS	The path to the NIRI public wavefront sensor components source directory	Not recommended
NIRI_WCSDIR	\${NIRI_DIR}/WCS	The path to the NIRI world coordinate system software directory.	Not recommended
NIRI_ISDIR	\${NIRI_DIR}/IS	The path to the NIRI instrument sequencer source directory	Not recommended
NIRI_DCDIR	\${NIRI_DIR}/DC	The path to the NIRI to GNAAC interface directory	Not recommended
NIRI_TARGETS	mv167	Which EPICS-supported hardware platform ^e .	Not recommended

- Items which are marked “Not recommended” will have unexpected results, unless the niri software is modified accordingly. Items which are marked “For debugging only” should be returned to the default value before running a working system. Details are outside the scope of this document.
- Note that this causes incorrect behavior, because local time is different from Gemini time.
- Note that this causes incorrect behavior, because local time is different from Gemini time.
- Note that this causes incorrect behavior, because the A&G status record will not be updated.
- The mv167 is the only supported architecture at this time.

You should not need to change any other files. If you do need to do so, it should be reported as a bug.

The csh/tcsh configuration file “niriLogin” sets up various environment variables and your path. You must source niriLogin before building or running any of the niri software. It would probably be wise to source “niriLogin” in your .login or .cshrc file:

```
source niriLogin
```

Run niriSetup to configure the software for your EPICS system, and copy in some standard EPICS files.

```
./niriSetup
```

6.5 Building the system

Execute the following command:

```
gmake
```

This should build the system. The software should compile without warnings, *except*

- All C modules will warn “‘rcsid’ defined but not used.”
- The SNL code compiler (*.stpp files) generates a lot of warnings about undefined variables.

6.6 Making dependency list

Switch to the source directory

```
cd $NIRI_DIR/src
```

Make the dependency list

```
gmake depends
```

Repeat with \$NIRI_LIBDIR/src, \$NIRI_ENGDIR/src, and \$NIRI_DCDIR/src.

This will insure that all files will be remade if the record configuration files or C header files are changed.

6.7 Loading the NIRI/CICS OIWFS/CICS control software or simulator

Connect to your IOC and set the start-up script to

- (...) /bin/mv167/startupSim (simulator),
- (...) /bin/mv167/startup (control software),
- (...) /WFS/bin/mv167/startupSim (OIWFS simulator), or
- (...) /WFS/bin/mv167/startup (OIWFS control software),

where (...) represents the full, absolute path to the niri_1.00 directory. Then reboot the IOC. The IOC should reboot without any errors. At this time, you should be running the software that you selected.

6.8 Running the user interface

The system may be started by running the command:

```
niriStart
```

This will start a top level control screen from which you can start the NIRI CC, WFS, DC, and low-level engineering screens. Note: This screen is designed to control the whole NIRI system. If it is used to control the OIWFS system, non-OIWFS control widgets will be white. Separate screens to control the OIWFS only are not provided.

6.9 Cleaning and rebuilding

The set up and installation process generates a number of files automatically. To insure that the old files are removed before building the system with a new configuration, type:

```
gmake reset
```

Do not use

```
gmake clean
```

because this fails to remove some files which should be removed, and remove some files which should not be removed. To restore the system to its original state, type:

```
gmake resetall.
```

7 Multiple copies of the software

The NIRI/OIWFS software is designed to make it easy for multiple copies of the software to be built and in use simultaneously on separate IOC's. Multiple copies may be kept in synch by using CVS. Each copy should be built in a separate directory and should have its own version of niriLogin, with the program prefix and name defined differently. In order to make it easy to have two copies simultaneously (i.e., a database which is intended for general use, and a version with experimental changes, or a simulator running simultaneously on another IOC, the following features are provided:

- All symbolic links are created by the configuration script, for ease of use with CVS.
- All configuration information is stored in niriLogin, and in files that are automatically generated by the configuration process.
- The only time that it is actually necessary to source niriLogin is before running niriSetup and the user interface scripts (niriEng, niriStart, etc.).
- It is not necessary to source niriLogin before running gmake.

8 Limitations of the simulation

The simulation is a reasonably accurate representation of the working niri system except:

- Mechanisms are idealized, i.e., all Hall-effect sensors lie on top of each other, there is no noise in the signals, there is no backlash, and the mechanisms cannot fail.
- The Hall effect sensors do not read the same values as the real mechanisms.
- The "FAST" simulation mode is not really very fast for datuming.

- The environment subsystem is a very poor simulation.

9 Unfinished tasks

This section describes tasks which remain to be done. These tasks, for the most part, are tasks that are fairly easy to do at the telescope, and very difficult to do with a standalone system.

9.1 WCS

1. Calibration constants for the WCS have not been computed.

This is easily done empirically, with far less effort than it would take to compute the values.

9.2 Networking and communications

1. The time bus has not been tested. Time functions have not been tested.

This requires equipment which is not available to the NIRI developers. The necessary subroutines are in place, and if they function as expected, this should be a simple operation.

2. The Interlock bus has not been tested.

NIRI currently does not need to generate an interlock event, nor will it need to respond to an interlock demand. The software and hardware are provided so that it will be possible to interface with the interlock bus if needed. This should be tested.

9.3 wfsBeam

1. The filter positions which block the light path to the wfs have yet to be entered in the appropriate .lut file.

This cannot be done until the final arrangements of filters is known.

10 Known bugs and limitations

This section describes problems with the NIRI software. All bugs listed here are believed to be non-critical or have a reasonable work-around. In general, bugs in the EPICS or GEM EPICS have not been fixed, if there is an adequate work-around, in the interest of portability. The problems here are generally of the following types:

- Non-fatal bugs which have proven difficult to isolate;
- Bugs which would require extensive redesign to repair;
- Bugs which would require significant time to repair, and which do not have a significant effect on the quality of the system;
- Bugs in the underlying EPICS or GEM EPICS system.

10.1 Problems with the simulation

1. The simulation starts in an invalid location.

When the system is first rebooted, its initial position is 0, and its initial Hall-effect sensor values are also 0.0. The Hall sensors will be computed correctly after the first motion command.

2. It is not possible to switch into simulation mode using the sequencer commands.

This was discussed with gemini, and it was decided that it was too difficult to implement with the existing tools. Simulation mode requires a separate startup script.

3. The GNAAC simulator is not supported.

The GNAAC simulator is very out of date and is not supported. The changes which are required to make it work with the NIRI software are probably small. The GNAAC simulator should be able to run on the same IOC as the NIRI software.

4. FAST Simulation mode is not really “FAST”

The FAST simulation mode takes a fair amount of time to complete some routine tasks (notably datuming). This could be improved with a small amount of effort, if the time and resources were made available.

5. VSM Simulation modes is not supported at all

The requirement for VSM was relaxed in discussions with Gemini. “FAST” mode is probably adequate for most purposes.

10.2 Problems with edd / dm

1. When the IOC is rebooted, some of the menus in the edd/dm user interface do not get loaded properly. The contents of the menu will be listed as “no menu.” Furthermore, some of the text / numeric displays show incorrect values (usually, obviously meaningless values).

This is a problem with edd/dm. A work-around is to close (not just iconify) the affected windows, and restart them.

2. The user interface dies unexpectedly.

This is a known bug with edd/dm. If you open a window (call it window A), and use a button on window A to open another window (call it window B), then close window A, then the user interface may crash unexpectedly at some time in the future. This is a known problem with edd/dm. There is no known work-around, other than to use caution when closing windows.

3. The user interface corrupts the color map of an X (including Openwindows) display.

This is a known bug with edd/dm. There is no known work-around. You will have to restart your windowing system.

4. The user interface screens cannot be edited with edd or displayed with dm.

All user interface tools for NIRI have been built using edd2.4 and dm2.4, which uses an incompatible file format. The following commands should be used in place of the edd and dm commands.

```
env EPICS_DISPLAY_PATH=`pwd`/O.solaris:`pwd` edd2.4
env EPICS_DISPLAY_PATH=`pwd`/O.solaris:`pwd` dm2.4
```

10.3 Stability problems

1. The system sometimes stops responding to EPICS channel-access commands (once per month).

This problem occurs randomly, and only when the cooling motors are in use. This may have been caused by a bug in the motor record, which has now been fixed.

10.4 Compiler Problems, Code, and Coding Standards

1. All C modules warn “‘rcsid’ defined but not used” during compilation.

The Gemini Programming Standards Document [22] requires “rcsid” to be defined and requires code to be compiled with a high warning level. These two requirements cannot be met simultaneously.

2. The SNL code compiler (*.stpp files) generates a lot of warnings about undefined variables.

These warnings are very difficult to suppress, because many constants are defined in C include files, but the SNL compiler cannot understand most C include files. Suppressing this error would require significant changes to header files, including system files, EPICS files, and Gemini-provided files or would require constants to be defined in more than one place, which is poor programming practice.

3. Boolean variables are not used consistently. Sometimes TRUE is 0, and sometimes it is any non-zero value. Sometimes FALSE is 1, and sometimes it is 0.

The EPICS Boolean convention (0 True, 1 False) is the reverse of the C convention (0 False, non-zero True). The EPICS convention is being used, the definitions EPICS_TRUE and EPICS_FALSE are always used. When using the C convention definitions are not always used. The engineering subsystem uses the C convention exclusively. All other systems should use the EPICS convention exclusively, except when interfacing to the Engineering subsystem, though there may be a few places where the C convention is used (these need to be corrected when found.)

10.5 Mechanical problems

1. The cooling motors do not stay synchronized.

There does not seem to be any way to fix this with the current hardware.

2. The software assumes that the number of motor-steps for a full rotation of one of the filter wheels and other wheel-like mechanisms is an integer.

Unfortunately, some of the wheels are a non-integer number of motor steps. This can cause slight positioning problems when the wheels have rotated a large number of turns from their home position. This is not believed to be a serious problem.

It may be necessary to datum the system occasionally or to use the diagnose command to return it to 0.

10.6 Datuming and diagnostic commands

1. The diagnose commands and redatum commands can be very slow for the wheels.

These commands move the wheels back to the zero point, without taking advantage of the coordinate wraparound. This could be improved, but the diagnose and redatum commands are intended for infrequent, engineering use.

2. If a mechanism is put into motion by a low-level engineering command, and if it is in the not-datumed state, then a warning message appears: "Mechanism position lost spontaneously."

It is not trivial to get rid of this warning message, because of the way that the SNL code is written. The warning message should just be ignored.

3. There is no warning if the pupil viewer is not properly seated.

The pupil viewer is an unusual mechanism. It seats itself by overrunning its limits and forcing itself into a hard stop. Handling this case would require a significant rewrite of the control code. In general, if the mechanism is ever handled manually, it is necessary to move it from one limit to the other or to datum it. (The high-level datum command automatically does a diagnose).

4. The wfsBeam record is of limited accuracy.

It would be difficult and time-consuming to figure out which orientations of the gimbal mirror would permit light to fall on the wave-front sensor with different pick-off mirrors. This and similar effects are completely neglected. The only things which are considered are mirror positioning and filter transparency.

10.7 Calibration tools

1. The scan tool times out a lot, especially for the focussing stage. When it times out, the user interface leaves most of the controls disabled. The interface must be exited and restarted to fix the problem.

The scan tool is used very infrequently, and never by the end users. This really *should* be fixed, but is a low priority.

2. The scan tool, the position tool, and the cycle tool handle don't handle text input very well.

The version of tcl / tk which is shipped with the GEM5 release is quite old. I do have a newer version of tcl / tk and have ported etwish to that release, but I don't use it with the

NIRI software, because that would make my utilities incompatible with the unmodified GEM5 release.

10.8 Other Problems

1. The number of frames for the observe command should really be part of the setup command.

The frames parameter is likely to change frequently, and shouldn't really require a complete set up of the detector controller in order to change its value. There could be a separate command, just to set the frame count.

2. Using caget or caput to access elements of type "double" in the HallStep record causes a fault, which renders the record non-functional.

There is probably some EPICS requirement that is not being satisfied by the HallStep record. Oddly enough, this problem only occurs with caget and caput. Edd / dm, cau, pvload, and etwish all have no problem.

3. There is no easy way, short of rebooting, to restore all parameters that can be set using the low-level engineering interface to their initial values. Using the low-level engineering interface, it is easy to put the system into a state in which the usual commands do not function properly, even though the system may appear normal. The low-level engineering interface is capable of putting the system into a state from which rebooting will not recover and can even put the system into a state from which software recovery is not possible.

There are just too many things that can be done with the low-level engineering interface to safeguard against all possibilities. There is no way to avoid this, except by disabling the low-level engineering commands completely. The low-level engineering commands are on screens which have a distinct black title bar with bright yellow letters. Be sure that you know what you are doing, before using the low-level engineering commands!

4. The current position for the guide probe can show "INVALID" when it should show "Undefined."

This is because of a limitation on the number of entries in a lutin/lutout record. In practice, the named positions are not used for the guide probe.

11 Possible Enhancements

The following modifications are candidates for release 1.01.

1. Datuming is too slow.

The datuming algorithm is very conservative, in order to work reliably with no modification under all conditions, both at room temperature and cryogenic temperatures.

2. The temperature control system is not as well automated as it should be.

Scheduling will make it impossible to complete the front end to the temperature control system. The existing code is sufficiently usable that simplifying the task is considered a low priority.

3. There are problems with the motor record.

Janet Tvedt discovered several problems with the motor record, and provided patches to fix these problems. These patches have not been incorporated into the hmotor record (NIRI’s customized motor record), and some of them should probably be applied. Currently, the hmotor record is working fairly reliably in the mode in which NIRI requires it, so there seems to be little need to apply these changes. If the hmotor record is used in other projects, however, there may be problems.

4. Utility scripts need to be rewritten.

The utility scripts in “scripts” should be rewritten in another language. They are really too complicated to be shell scripts. In particular, there are a number of global variables which are reused everywhere. It is not a serious problem with the existing scripts, but requires careful attention to which variables are reused.

5. Health problems cannot be traced to alarms

This is a Gemini requirement which is not properly met. In fact, although the health tree is quite extensive, alarms are practically non-existent.

6. Issuing multiple movement commands to a single mechanism can cause problems. In particular, it is not possible to issue a datum command and a motion command at the same time.

There is a race condition, which can occur. It would be possible to add an interlock to keep this from happening, but for now, nothing is done. This should never occur in normal operation anyway; however, it should really be fixed.

12 Design Decisions (“It’s not a bug, it’s a feature!”)

1. Mechanisms do not datum themselves automatically.

Mechanisms that datum themselves automatically can be a real pain to work with during hardware and software testing and integration. Furthermore, the mechanisms retain knowledge of their current positions, even through a reboot (though not a power cycle), which can be a significant time saver, allowing the entire time-consuming datuming procedure to be bypassed. Since NIRI does not appear to need frequent rebooting under normal conditions, this is not a serious problem.

2. The Health state is not set to the WARNING mode if an operation fails.

The CICS sets its health to WARNING if an operation fails. This seemed to be rather unintuitive and was misleading to the people who tested the NIRI software, so it was removed. In general, any failure which can cause an operation to fail will trigger some warning, most commonly, the backlash warning or the not-datumed warning.

3. If a mechanism times out, the CAR record is not set to the ERR state.

Timeout is now handed via a warning SIR record, instead of via the CAR record. This is because certain operations can take a very long time under certain conditions. For example, datuming all mechanisms is much more time consuming than might be expected, because all mechanisms cannot be operated simultaneously.

